

Data Science Independent Study

Machine Learning Project Report

July 31, 2026

Introduction:

Snowfall is an important component of Colorado's climate, affecting transportation, water resources, recreation, and communities across the state. Snowfall amounts can vary considerably by location and time of year, making historical weather observations a valuable source of data for examining seasonal and geographic patterns and developing predictive models. Machine learning provides an opportunity to identify patterns within these historical observations and evaluate how effectively they can be used to predict daily snowfall. The objective of this project was to develop a supervised machine learning model for predicting daily snowfall amounts across Colorado using historical weather station observations obtained directly from the National Oceanic and Atmospheric Administration (NOAA). Data were collected through the NOAA National Centers for Environmental Information (NCEI) Climate Data Online API using the Global Historical Climatology Network Daily (GHCN-Daily) dataset. Following an evaluation of station reporting coverage, the analysis included 245 Colorado weather stations and approximately 747,000 snowfall observations from 2016 through 2024.

The project demonstrates an end-to-end machine learning workflow consisting of data acquisition, data cleaning and preparation, exploratory data analysis, feature engineering, model development, hyperparameter tuning, and performance evaluation. Exploratory analysis identified strong seasonal and geographic differences in snowfall. A Random Forest regression model was subsequently selected because of its ability to capture nonlinear relationships and interactions among temporal and location-based features without requiring assumptions about the underlying distribution of snowfall. Random Forest was selected because of its ability to model nonlinear relationships and interactions among temporal and location-based features without requiring assumptions about the underlying distribution of snowfall. Model performance was evaluated against a baseline model and subsequently improved through hyperparameter optimization.

Data Source & Preparation

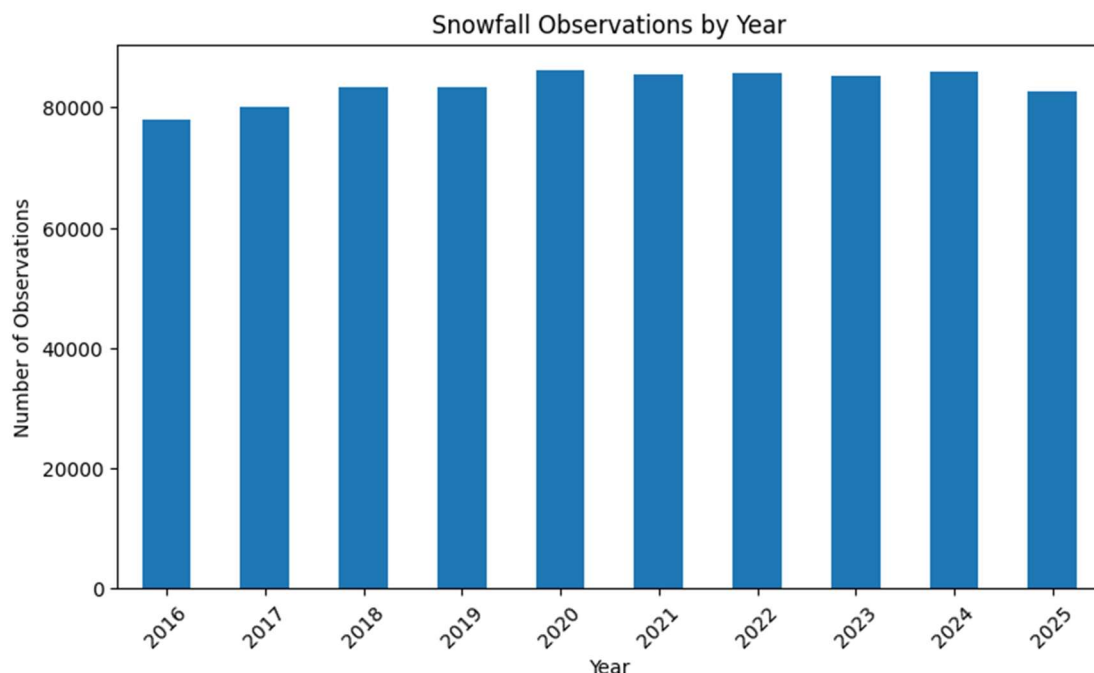
The dataset for this project was obtained from the National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (NCEI) Climate Data Online API. The project uses observations from the Global Historical Climatology Network Daily (GHCN-Daily) dataset, which provides daily weather observations collected from monitoring stations. The initial data pull was restricted to Colorado and was used to evaluate the structure, availability, and coverage of snowfall data before constructing the larger machine learning dataset. The initial API response was returned in long format, with each row representing an individual weather observation. Available fields included the observation date,

weather measurement type, weather station identifier, NOAA observation attributes, and recorded numeric value. Initial exploration identified several available weather measurements, including precipitation (PRCP), snowfall (SNOW), snow depth (SNWD), and water-equivalent snow measurements. Because the objective of the project was to predict daily snowfall amounts, SNOW was selected as the target variable.

To construct the historical dataset, snowfall observations were collected for the period from January 1, 2016, through December 31, 2024. Weather stations were evaluated based on the completeness of their snowfall reporting, and stations with insufficient historical coverage were removed. Following this filtering process, 245 Colorado weather stations were retained for the modeling dataset. The data was also reviewed for missing and malformed records. The final cleaned dataset contained approximately 747,000 daily snowfall observations.

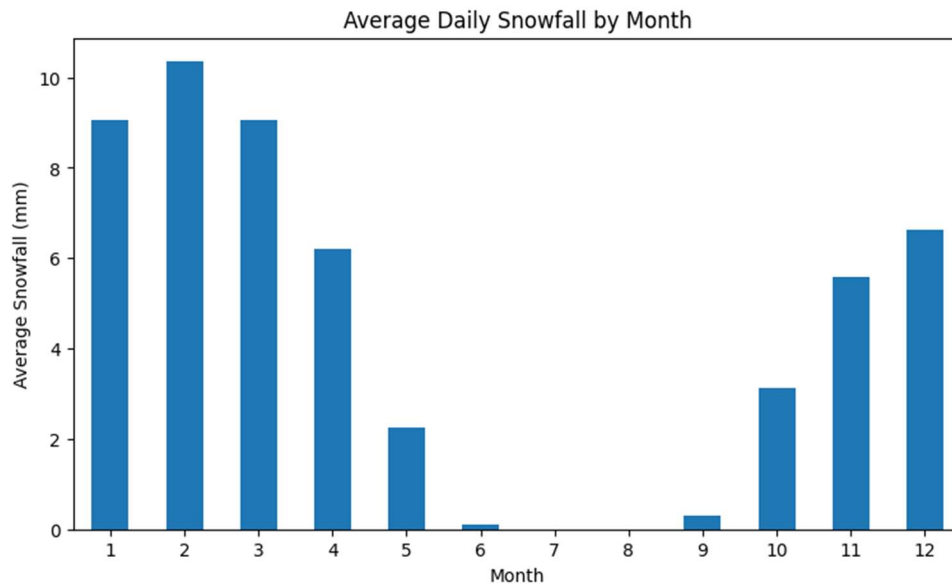
Exploratory Data Analysis

Exploratory data analysis was conducted in stages, beginning with an examination of the structure and availability of the NOAA data and progressing to an analysis of the final historical snowfall dataset. An initial sample of 1,000 Colorado weather observations was retrieved from the NOAA API and contained data from 400 unique weather stations. The sample included precipitation (PRCP), snowfall (SNOW), snow depth (SNWD), and other weather measurements. Subsequent analysis was restricted to SNOW observations.

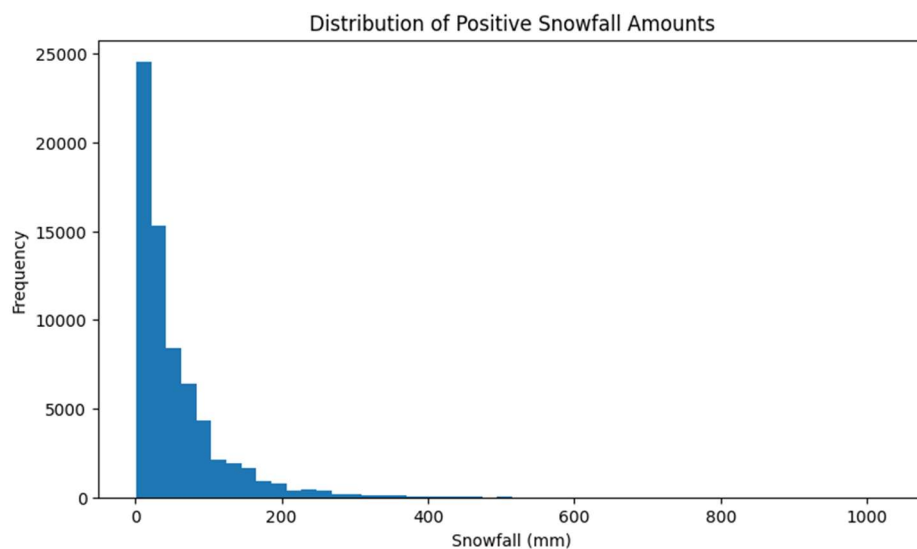


The initial exploration also identified an important limitation of the NOAA API. Although the request covered multiple dates, the 1,000 returned observations were all from a single day because the API reached its 1,000-record request limit before progressing to subsequent dates. As

a result, the historical dataset could not be retrieved through a single API request and required multiple requests to collect complete observations across stations and years.



To identify stations with sufficient data for modeling, Colorado snowfall observations for 2024 were analyzed based on both total reported snowfall and the number of observations recorded by each station. Observation frequency was considered important because selecting stations based only on snowfall totals could favor locations with incomplete reporting. Stations reporting at least 300 snowfall observations during 2024 were initially retained, resulting in 382 candidate stations.



Historical snowfall observations were then collected for the selected stations across the study period. Because of NOAA API record and request-length limitations, stations were divided into

smaller groups and the data was retrieved through multiple paginated requests before being combined into a single dataset. Station coverage was evaluated again across the complete historical period, and stations with insufficient reporting were removed. This process resulted in a final dataset containing approximately 747,000 observations from 245 Colorado weather stations covering 2016–2024. The final dataset showed clear seasonal and geographic patterns in snowfall. Average snowfall was highest during the winter and early spring months and lowest during the summer, while total snowfall varied considerably among weather stations. The distribution of positive snowfall amounts was strongly right-skewed, with smaller snowfall events occurring much more frequently than extreme events. Additionally, 91.6% of observations recorded no snowfall and 8.4% recorded measurable snowfall, identifying substantial target imbalance as an important consideration for subsequent model development.

Feature Engineering

Following data cleaning and analysis, several features were engineered from the observation date to represent the temporal and seasonal patterns associated with snowfall. These features included year, month, day of month, week of year, day of year, day of week, and meteorological season (Winter, Spring, Summer, and Fall). The weather station identifier was also retained as a categorical feature to account for geographic differences in snowfall across Colorado.

For model preparation, the features were separated into numeric and categorical variables. Numeric temporal features were passed directly into the model, while categorical variables, including weather station and season, were transformed using One-Hot Encoding. These preprocessing steps were incorporated into a Scikit-learn pipeline to ensure that the same transformations were consistently applied during model training and prediction. This process transformed the cleaned historical observations into a structured feature set that could be used to train the Random Forest regression model. These features were selected to represent the seasonal and geographic patterns identified during exploratory analysis while maintaining a relatively interpretable feature set.

Model Development and Evaluation

The cleaned dataset was formulated as a supervised regression problem in which daily snowfall amount served as the target variable. The data were randomly divided into 80% training and 20% testing subsets. For training the Random Forest, a random sample of 100,000 observations from the training set was used for initial model development and hyperparameter tuning. Model performance was evaluated on the held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Three modeling stages were used to evaluate the ability of the available temporal and location-based features to predict daily snowfall. A simple baseline model was first established to provide a reference point for evaluating whether the machine learning model learned meaningful

patterns. An initial Random Forest regression model was then trained and evaluated, followed by a tuned Random Forest model developed through hyperparameter optimization.

Baseline Model: A baseline regression model was established by predicting the mean snowfall value from the training data for all observations in the test set. This model did not attempt to learn relationships between the predictor variables and snowfall, but it provided a minimum performance benchmark for comparison with the Random Forest models. The baseline produced a Mean Absolute Error (MAE) of 8.61 mm, a Root Mean Squared Error (RMSE) of 24.21 mm, and an R^2 value of approximately 0.000. The R^2 value indicates that predicting the historical mean did not explain variation in daily snowfall.

Initial Random Forest Model: A Random Forest Regressor was selected as the primary machine learning algorithm because it can capture nonlinear relationships and interactions among temporal and geographic variables. The model was incorporated into a Scikit-learn pipeline that passed numeric variables directly to the estimator while applying One-Hot Encoding to the categorical station and season variables. The initial Random Forest used 25 trees, a maximum depth of 15, a minimum leaf size of 5, square-root feature sampling, and parallel processing. The initial Random Forest achieved an MAE of 7.93 mm, RMSE of 23.52 mm, and R^2 of 0.0562. Compared with the mean baseline, the Random Forest reduced both MAE and RMSE and explained approximately 5.6% of the variation in daily snowfall. Although the improvement was modest, the results demonstrated that the temporal and station-based features contained predictive information beyond the historical average.

Feature Importance: Feature importance analysis showed that the model relied primarily on seasonal timing and station location. Day of year accounted for approximately 20.8% of total feature importance, followed by weather station at 20.2%. Week of year, month, day of month, and season also contributed to predictions, indicating that the model learned Colorado's seasonal and geographic snowfall patterns. Several temporal predictors, including day of year, week of year, month, and season, represent related aspects of annual seasonality and should not necessarily be interpreted as independent effects.

Hyperparameter Tuning: The Random Forest was optimized using RandomizedSearchCV. The search evaluated different values for the number of trees, maximum tree depth, minimum samples required to split a node, minimum samples per leaf, and the number of features considered at each split. Three-fold cross-validation was used to evaluate 10 randomly selected parameter combinations, resulting in 30 model fits. RMSE was used as the optimization criterion.

Tuned Random Forest: The best-performing configuration used 25 trees, no maximum depth restriction, a minimum of 10 samples required to split a node, a minimum leaf size of

2, and square-root feature sampling. The optimized configuration achieved a cross-validation RMSE of approximately 23.08 mm. When evaluated against the held-out test data, the tuned Random Forest achieved an MAE of 6.90 mm, RMSE of 22.07 mm, and R^2 of 0.1691. Hyperparameter tuning therefore produced a meaningful improvement over the initial Random Forest. The MAE decreased by approximately 13.0%, RMSE decreased by approximately 6.2%, and the proportion of explained variance increased from approximately 5.6% to 16.9%.

Table 1. Comparison of Regression Model Performance

Model	MAE (mm) ↓	RMSE (mm) ↓	R^2 ↑
Mean Baseline	8.61	24.21	0.0000
Initial Random Forest	7.93	23.52	0.0562
Tuned Random Forest	6.90	22.07	0.1691

Results and Discussion

The results demonstrate a clear progression in predictive performance. The initial Random Forest outperformed the mean baseline, showing that station location and temporal features provided useful information for predicting snowfall. Hyperparameter tuning produced a further improvement, with the final model reducing MAE from 7.93 mm to 6.90 mm and increasing R^2 from 0.0562 to 0.1691. The tuned Random Forest was therefore selected as the final model because it achieved the lowest prediction errors and explained the greatest proportion of variation in daily snowfall.

The use of a random train-test split also represents a limitation of the current analysis because observations from multiple years were included in both the training and testing sets. Future analysis could use a chronological split in which earlier years are used for training and a later year is reserved for testing. This approach would provide a more realistic evaluation of the model's ability to predict snowfall for future time periods.

Although the tuned model demonstrated meaningful improvement, an R^2 of 0.1691 indicates that approximately 83% of the variation in daily snowfall remains unexplained by the current feature set. This finding suggests that the model successfully captured broad seasonal and geographic patterns but lacked the meteorological information necessary to explain individual snowfall events.

Conclusion

This project demonstrated an end-to-end machine learning workflow for predicting daily snowfall across Colorado using historical observations obtained directly from the NOAA NCEI

API. The analysis progressed from API-based data acquisition and station coverage evaluation through exploratory analysis, feature engineering, Random Forest regression, feature importance analysis, and hyperparameter optimization. The final dataset contained approximately 747,000 observations from 245 Colorado weather stations covering 2016 through 2024.

The tuned Random Forest produced the strongest performance, achieving an MAE of 6.90 mm, RMSE of 22.07 mm, and R^2 of 0.1691. These results demonstrate that temporal and geographic information provides meaningful predictive value, particularly through day of year and weather station location. However, much of the variability in daily snowfall remained unexplained, highlighting the limitations of predicting individual snowfall events using calendar and location information alone. Future work could extend the model by incorporating meteorological predictors such as precipitation and temperature, evaluating time-based validation approaches, and comparing Random Forest performance with additional regression algorithms.